

HINTS AND SOLUTIONS

Q.1 B

Sol. No. of field lines from a charge $\propto$ magnitude of charge

Q.2 D

Sol. $E = -\frac{dV}{dr} = a$; So flux $-4\pi R^2 a$

$$\text{Now } \frac{q_{\text{in}}}{\epsilon_0} = 4\pi R^2 a \Rightarrow q_{\text{in}} = 4\pi R^2 \epsilon_0 a$$

Q.3 C

Q.4 B

Q.5 D

Q.6 A

Q.7 B

Q.8 B

Sol. $|V_A - V_B| = kq \left(\frac{1}{R} - \frac{1}{r} \right) = \frac{9 \times 10^9 \times 0.25 \times 10^{-9}}{10^{-2}} \left(\frac{1}{5} - \frac{1}{15} \right) = 30 \text{ V}$

Q.9 C

Sol. WD by $\vec{F} = \vec{C} \times \vec{V}$ will be zero

$$\text{since } WD = \int \vec{f} \cdot d\vec{r} = \int \vec{f} \cdot \vec{v} dt = \int (\vec{C} \times \vec{V}) \cdot \vec{V} dt = 0$$

$$WD = \Delta K + \Delta U$$

$$0 = \left[\frac{1}{2} m \left(\frac{v_0}{2} \right)^2 - \frac{1}{2} m v_0^2 \right] + qEd$$

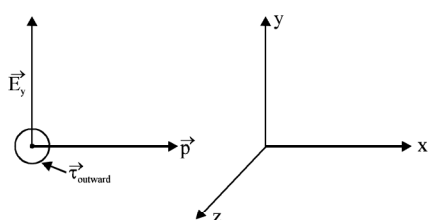
Q.10 B

Sol. $\vec{F} = p \frac{\partial \vec{E}}{\partial \ell}$

$$\text{Force is directed along y-axis as } \left| \frac{\partial \vec{E}}{\partial \ell} \right| = \frac{\partial E_y}{\partial x} \neq 0$$

Q.11 C

Sol. $\vec{\tau} = \vec{p} \times \vec{E}$



Torque is due E_y therefore it points along z-axis.]

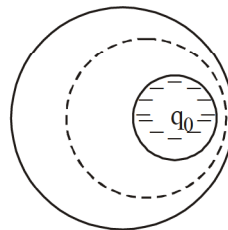
Q.12 A

Sol. Since $E_{\text{inside conductor}} = 0$

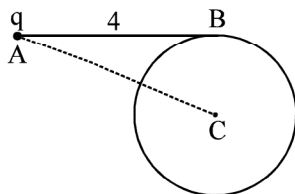
$$\oint \vec{E} \cdot d\vec{s} \text{ over gaussian surface} = 0$$

$$\Rightarrow \oint \vec{E} \cdot d\vec{s} = \frac{q_{\text{in}}}{\epsilon_0} = \frac{q_0 + q_{\text{in on surface}}}{\epsilon_0}$$

$$\Rightarrow q_{\text{in on surface}} = -q_0$$



Q.13 AC



Sol.

$$AC = 5 \text{ m}$$

$$V = \frac{kq}{AC} = \frac{9 \times 10^9 \times 1 \times 10^{-6}}{5} = 1.8 \times 10^3 = 1.8 \text{ kV}$$

$$V_B = (V_B)_{\text{due to } q} + (V_B)_i$$

$(V_B)_i$ = Potential at B due to induced charge

$$\therefore 1.8 \times 10^3 = \frac{kq}{AB} + (V_B)_i$$

$$1.8 \times 10^3 = 2.25 \times 10^3 + (V_B)_i$$

$$(V_B)_i = -0.45 \text{ kV}$$

So, (A) & (C) are correct

Q.14 BC

Q.15 BCD

[Ans. (e) through (h)]

Q.16 ABD

Sol. By using work energy theorem

Q.17 [Ans. (A) P, R, S (B) P, Q, R, S, T (C) Q, T (D) R, S]

Q.18 [Ans. 1.8]

$$\text{Sol. } U_i = \frac{kQ^2}{2R} + \frac{k \cdot 3Q}{2R} \times (-2Q) + \frac{k(2Q)^2}{2 \times 2R} + \frac{2kQ}{R} \times Q$$

$$\frac{kQ^2}{R} \left[\frac{1}{2} - 3 + 1 + 2 \right] = \frac{kQ^2}{2R}$$

$$\frac{kQ^2}{2R} - \frac{k2Q}{R}(-Q) - \frac{kQ}{2R}(-2Q) - \frac{k(2Q)^2}{2 \times 2R}$$

$$\frac{kQ^2}{R} \left[\frac{1}{2} - 2 - 1 + 1 \right]$$

$$\frac{3kQ^2}{2R}$$

$$\frac{2kQ^2}{R}$$



Q.19 Ans $\frac{3\sigma\lambda}{2m\epsilon_0}$

Q.20 [Ans. 0001]

Sol. $\oint \vec{E} \cdot d\vec{A} = \frac{\int \rho dV}{\epsilon_0}$

$$E \times 4\pi r^2 = \frac{k \int r^n \times 4\pi r^2 dr}{\epsilon_0} = \frac{4\pi k}{\epsilon_0} \frac{r^{n+3}}{n+3}$$

$$E = \frac{k}{(n+3)\epsilon_0} (r^{n+1})$$

$$n+1=2 \Rightarrow n=1$$

Q.21 [Ans. 2]

Sol. $\frac{\phi}{\phi_T} = \frac{\Omega}{4\pi} = \frac{2\pi}{4\pi} (1 - \cos \theta)$

$$\tan \theta = \frac{4}{3} \quad \phi = \frac{q}{2\epsilon_0} \left(1 - \frac{3}{5}\right)$$

$$\theta = 53^\circ \quad \frac{q}{5\epsilon_0}$$

$$10n=2$$

